

HEALTHCONNECT CLINIC

Appointment No-Show Analysis

Week 8 — Final Analytics & Decision Support Package

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Data Analytics Track

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Presentation Outline

1 The Problem

High no-show rates causing operational and financial pressure at HealthConnect Clinic

2 5 KPI Results

Validated metrics from Python and SQL analysis of 5,000 appointment records

3 Advanced Findings

Deep dives into December patterns, High Risk patients, and waiting time

4 Integration & Recommendations

Cross-track collaboration, HC-POD record, business recommendations

SECTION 1 — THE PROBLEM

Nearly half of all appointments at HealthConnect are missed.

48.5%

Overall No-Show Rate

2,423 of 5,000 appointments missed

- ⚠ Empty consulting rooms and wasted clinical resources every single day
- 👤 Patients who need care are not receiving it — attendance just 46.3%
- 📊 The pattern is predictable — which means it is solvable with data

Central question: How can HealthConnect use data and AI to reduce missed appointments?

The Dataset

5,000
Appointment Records

18
Variables per record

0
Duplicate records found

Key columns analysed:

Column	Purpose
booking_lead_days	Days between booking and appointment
previous_no_shows	Patient risk history
reminder_sent / channel	SMS, phone, or email reminder
appointment_type	General, Follow-up, Specialist, Diagnostic
appointment_outcome	Target: No-Show / Attended / Cancelled

Two engineered features added: **lead_time_band** (categorical) and **patient_risk_tier** (Low / Medium / High)

SECTION 2 — KPI RESULTS

5 Validated KPIs — All 8 Tests Passed

KPI 1

Overall No-Show Rate

48.46%

KPI 2

Reminder Effectiveness

4.03 pp

KPI 3

Lead Time Effect

27.81% → 60.49%

KPI 4

Risk Tier Rates

43% → 61%

KPI 5

No-Show by Appointment Type

Follow-up 51.23% · Specialist 47.44% · General 46.64% · Diagnostic 49.75%

KPI 1

Overall No-Show Rate

48.46%

No-Show Rate

$2,423 \div 5,000 \times 100$

Formula

$(\text{No-Shows} \div \text{Total Appointments})$
 $\times 100$

Outcome	Count	Rate
No-Show	2,423	48.46%
Attended	2,314	46.28%
Cancelled	263	5.26%
Total	5,000	100%

Business impact: For every 100 appointments scheduled, only 46 are attended. Empty consulting rooms cost the clinic both revenue and patient trust.

Test T1 — Verified: $2,423 \div 5,000 \times 100 = 48.46\%$ ✓

KPI 2

Reminder Effectiveness Rate

4.03 pp

Reduction in no-shows

51.39% – 47.36%

By channel (where sent):

SMS: **47.31%** | Phone: **47.51%** |

Email: **47.38%**

Reminder Status	Total	No-Shows	Rate
No Reminder Sent	1,366	702	51.39%
Reminder Sent	3,634	1,721	47.36%

Insight: Reminders work, but the 4.03 pp reduction is modest. The opportunity is in **targeting high-risk patients** with the right channel at the right time — not just sending more reminders.

Test T2 — Verified: 51.39% – 47.36% = 4.03 pp ✓

KPI 3

No-Show Rate by Booking Lead Time

The longer patients wait, the more they default

0–7 days **27.81%**

8–14 days **33.55%**

15–30 days **43.21%**

31–60 days **60.49%**

The 2.2× effect

Patients who book 31–60 days out are **2.17 times more likely** to miss their appointment than those who book within the same week.

Recommendation:

Send additional reminders closer to the appointment date for patients who booked more than 14 days in advance.

Test T3 — All 4 rates verified independently ✓

KPI 4

No-Show Rate by Patient Risk Tier

● Low Risk (0 prior no-shows)

43.51%

2,736 patients

● Medium Risk (1 prior no-show)

53.49%

1,733 patients

● High Risk (2+ prior no-shows)

61.02%

531 patients

The 17.51 pp gap

High Risk patients miss at a rate 17.51 percentage points higher than Low Risk patients — a clear, predictable pattern.

Feature used by Data Science:

The **patient_risk_tier** engineered feature improved ML model precision and recall for High Risk patients.

Test T4 — Confirmed: Low < Medium < High ✓

KPI 5

No-Show Rate by Appointment Type

Follow-up

Highest miss rate

51.23%

Diagnostic Test

Second highest

49.75%

Specialist

Below average

47.44%

General

Lowest miss rate

46.64%

Weighted average check

The type-weighted average equals exactly 48.46% — matching KPI 1 precisely, confirming arithmetic consistency.

Why follow-ups are highest:

Patients may feel better after a first visit and de-prioritise the follow-up — perceived urgency is lower.

Test T5 — Weighted avg = 48.46% ✓

Three Refinements from Week 7

Week 7 identified gaps in the Week 6 analysis and resolved each one with targeted deep-dive investigations.

R1

December Seasonal Spike

Why was December 2025 the worst month at 53.85%? Investigation revealed which appointment types drive the spike.

R2

High Risk Best-Case Benchmark

Under optimal conditions (SMS + same-week booking), what is the lowest achievable no-show rate for High Risk patients?

R3

Waiting Time as Patient Experience

Waiting time data was never used analytically — Week 7 applied it as an operational quality metric.

R2 High Risk Best-Case Benchmark

What is the lowest achievable no-show rate for High Risk patients under the best possible conditions?

Condition	Count	No-Show Rate
No reminder sent	190	68.24%
Overall High Risk	531	61.02%
High Risk + SMS reminder	192	56.25%
High Risk + SMS + 0–7 day booking	29	37.93%

30.31 pp improvement over the no-reminder baseline. Same-week SMS reminders are the single most effective combination for this patient group.

Intervention Target

37.93%

Adopted as the ML model
evaluation benchmark

This benchmark was shared with the Data Science track as the target rate the model must beat to prove clinical value.

R1 December 2025 Seasonal Breakdown

December had a 53.85% no-show rate — the worst month. **Which appointment types drove this spike?**

Appointment Type	December Rate	Overall Rate	Difference
Follow-up	56.63%	51.23%	+5.40 pp
Specialist	55.10%	47.44%	+7.66 pp
General	53.40%	46.64%	+6.76 pp
Diagnostic Test	44.00%	49.75%	−5.75 pp

Key finding

Specialist Consultations show the biggest December spike (+7.66 pp). Diagnostic Tests actually improve in December — possibly due to more urgent, non-deferrable procedures.

This breakdown was shared with the Data Science track as a potential seasonal model feature.

R3

Waiting Time – Patient Experience Metric

24.29

minutes average wait
across all appointment types

By appointment type:

Follow-up: **24.77 min** (longest)

General: **24.35 min**

Specialist: **24.20 min**

Diagnostic: **23.85 min**

By day of week:

Tuesday (busiest)

25.34 min

Monday

24.88 min

Wednesday

24.41 min

Friday (quickest)

22.48 min

Note: Waiting time is only recorded for attended appointments. It cannot predict no-shows but serves as a direct measure of patient experience quality.

SECTION 4 — VALIDATION & INTEGRATION

8 Formal Tests

All 8 Passed ✓

Test	What Was Tested	Verification	Result
T1	KPI 1 — Overall No-Show Rate	$2423 \div 5000 = 48.46\%$	PASS
T2	KPI 2 — Reminder Effectiveness	$51.39\% - 47.36\% = 4.03 \text{ pp}$	PASS
T3	KPI 3 — Lead Time Band Rates	27.81% to 60.49%	PASS
T4	KPI 4 — Risk Tier Ordering	$43.51\% < 53.49\% < 61.02\%$	PASS
T5	KPI 5 — Weighted Average Check	Type avg = 48.46% ✓	PASS
T6	Age group — two independent methods	All 6 age groups matched exactly	PASS
T7	Reminder channel null integrity	0 invalid null violations found	PASS
T8	Lead time band completeness	$640+602+1333+2425 = 5,000$	PASS

All KPI arithmetic independently verified — no discrepancies found across Weeks 5, 6, and 7

Cross-Track Integration — Data Science

PROVIDED TO DATA SCIENCE ↑

- KPIs** 5 validated KPI definitions and verified figures as feature engineering targets
- F1** lead_time_band — categorical feature (0–7 / 8–14 / 15–30 / 31–60 days)
- F2** patient_risk_tier — ordinal feature (Low / Medium / High based on previous no-shows)
- BM** 37.93% best-case benchmark — adopted as the ML model evaluation target
- S1** December seasonal breakdown — flagged as a potential model input for month-of-year seasonality

RECEIVED FROM DATA SCIENCE ↓

Confirmation that **engineered features improved both precision and recall** for the High Risk patient segment in the predictive model

lead_time_band and **patient_risk_tier** ranked among the top predictive features in their initial model

The 37.93% benchmark was **incorporated into model evaluation criteria** across the HC-POD

Impact: Analytics directly shaped the features the ML model was trained on — a clear pipeline from descriptive insight to predictive capability.

HC-POD Integration Record

How the Analytics track connected to every other track in the pod

Track	What We Provided	What We Received	Integration Outcome
Data Science	Validated KPIs, 2 engineered features, 37.93% benchmark	Feature importance confirmation, model evaluation framework	Engineered features ranked top predictors; benchmark adopted pod-wide
ML Engineering	Feature definitions, risk tier logic, December seasonal flag	Model precision/recall metrics for High Risk segment	Analytics features embedded in the deployed model pipeline
GenAI	Business context on no-show patterns, KPI descriptions	Natural-language insight summaries for the HealthConnect app	Analytics findings surfaced as patient-facing messaging in the app
Project Mgmt	Weekly KPI updates, testing evidence, cross-track dependency log	Sprint alignment, deadline coordination, integration sign-off	All Analytics deliverables delivered on schedule across Weeks 5–8
Business Analyst	5 recommendations grounded in KPI evidence	Stakeholder requirement alignment, business framing	Recommendations formatted for clinic operations audience

Bottom line: The Data Analytics track served as the evidential foundation for every other HC-POD track — every model, every app feature, and every business recommendation traced back to the validated KPI framework.

5 Business Recommendations

R1

Prioritise Same-Week SMS for High Risk Patients

Target 37.93% no-show rate by combining SMS reminders with 0–7 day booking windows for patients with ≥ 2 previous no-shows. Current High Risk baseline is 61.02%.

R2

Re-engagement Protocol for Far-Advance Bookings

Appointments booked 31–60 days out have a 60.49% no-show rate. Introduce a confirmation re-contact at the 7-day mark to reduce late cancellations and silent non-attendance.

R3

December Seasonal Intervention

Deploy additional outreach in November–December targeting Follow-up (+5.40 pp) and Specialist (+7.66 pp) appointment types. Protect Diagnostic Test capacity, which actually improves in December.

R4

Reduce Tuesday Waiting Times

Tuesday averages 25.34 minutes — the longest day. Stagger appointment slots or add triage capacity on Tuesdays to bring waiting time closer to the Friday benchmark of 22.48 minutes.

R5

Integrate ML Model Using Validated Features

Deploy the predictive model built by the Data Science track using lead_time_band and patient_risk_tier as primary features. Use model outputs to auto-prioritise reminder dispatch. Target: beat the 37.93% analytical benchmark to prove clinical value.

Analytical Limitations

Acknowledged constraints on the scope and generalisability of this analysis

No qualitative data

The dataset contains no information on why patients miss appointments — no cancellation reasons, no patient surveys, no call logs. All findings are behavioural correlations, not causal explanations.

Waiting time is not predictive

Waiting time is only recorded for attended appointments. It cannot be used to predict no-shows and was analysed solely as a patient experience metric.

Small benchmark sample (n = 29)

The 37.93% best-case rate is derived from 29 High Risk patients who received an SMS and booked within 7 days. Statistical confidence at this sample size is limited — the benchmark is directional, not definitive.

Fictional synthetic dataset

This dataset was created for the AnalystLab internship programme. Findings are analytically valid within the dataset but cannot be applied directly to real-world clinic operations without validation on live data.

Cross-track coordination assumed

Cross-track outputs (e.g. model precision/recall improvements from the Data Science track) were reported by collaborating tracks. Independent verification of their figures was outside the scope of this track.

No control group for interventions

Reminder effectiveness is measured by comparing reminded vs. non-reminded patients within the same dataset — not through a randomised experiment. Selection effects (e.g. who was sent reminders) may influence results.

Professional Reflection

Eight weeks of growth — from raw data to decision-ready insight

TECHNICAL GROWTH

Moved from basic Excel thinking to Python-first analysis with pandas. Learned to engineer features from raw columns, build reusable functions, and verify outputs through multiple independent methods.

CROSS-TRACK COLLABORATION

Sharing the 37.93% benchmark with the Data Science track and seeing it adopted pod-wide showed me that analytics is most valuable when it is designed to be used — not just documented.

BIGGEST CHALLENGE

The December surprise — Diagnostic Tests improving while all other types worsened — reminded me to let the data lead, not my assumptions. I almost missed it by expecting uniformity.

ANALYTICAL RIGOUR

The 8 formal validation tests taught me that correctness is not enough — you have to prove correctness. I learned to document every figure and build an audit trail that any reviewer can follow.

COMMUNICATION

Writing weekly reports forced me to translate statistical findings into business language. The discipline of "so what?" — always ending every finding with its implication — changed how I think about presenting data.

GOING FORWARD

I want to deepen my machine learning knowledge so I can move beyond describing what happened and start building systems that predict and prevent it.

**From 48.46% no-show rate
to a 37.93% target —
evidence-led, pod-wide.**

5

Validated KPIs

8

Tests Passed

3

Advanced Deep Dives

5

Recommendations

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